Sunbeam Infotech

Exploring new ideas, Reaching new heights!

Database

Data storage & manipulation
→ File IO
→ Data mngmt system
① Excel, DB Star, → File based db
② DBMS → CRUD ops
Ⓐ RDBMS → Relational DBMS
Ⓑ NoSQL → Non-Relational DBMS

RDBMS → Relational DBMS } Based on Set theory
Relational = Tabular. } - E.F. Codd rules
- 1970

Students		
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Courses		
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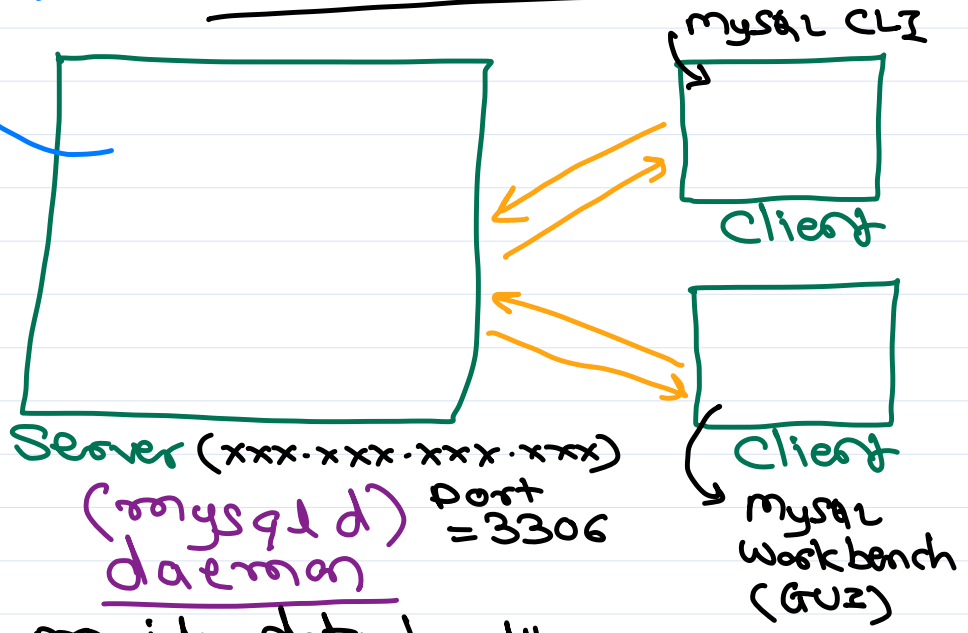
Relation

RDBMS
① Oracle
② MySQL
③ PostgreSQL
④ MS-SQL Server

Internally stores data on hard disk →
C:\ProgramData\MySQL\MySQL Server\... MySQL - RDBMS

C - Create (Insert)
R - Retrieve (Select)
U - Update
D - Delete



provide data handling service over network (socket = ip addr + port)

DB cmds are written in SQL language. Cmds → Queries.

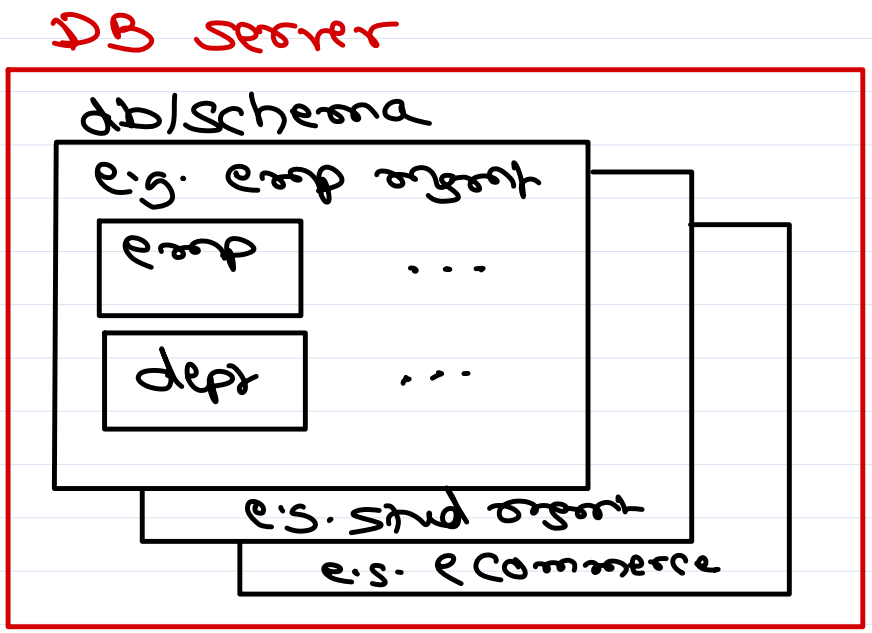
SQL = Structured Query Language

ANSI standardized in 1987.

Standard revised multiple times.

Refer: wiki → SQL.

1. DDL → Data Defn
   → CREATE, DROP, TRUNCATE, ALTER
2. DML → Data Manipulation
   → INSERT, UPDATE, DELETE
3. DQL → Data Query → SELECT
4. DCL → Data Control
   → GRANT, REVOKE
5. TCL → Tx Control
   → COMMIT, ROLLBACK



Thank you!

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